



Lab 1: Modelling of the Motor System

MTE 484: Digital Control Systems

Instructor Sergio Livera

Department of Mechatronics and Mechanical Engineering

Levi Rogalla
Alex Veric

October 2, 2026

1 Introduction

This lab introduces the motor and gear mechanism which are to be used for the 'ball balancing on beam' project, which must be properly characterized in order to facilitate the completion of the project. This includes experimentally determining values for 'stiction' and the transfer function of the motor. These activities were performed during the span of the lab session, and their procedures and results are described in detail in the following report.

2 Question C

By this point in the lab, the set up was already complete. The Arduino was functioning as intended, and with the sample code provided, it was possible to both send voltage values to the motor and read sensor data, which included the motor rotational position and the ball's position on the beam. These values were printed out on the serial monitor, allowing them to be recorded.

The values of the motor position being output needed to be converted to radians in order to be useful. Using the following formula, the motor position output values could be linearized and converted into radians:

$$p_{radians} = m * p_{sensor} + b \quad (1)$$

Where the m and b values must be determined through a system of equations, using experimental values of p_{rad} and p_{sensor} to do so.

The position of the motor was manually changed to three points: 0, $\frac{\pi}{4}$, and $-\frac{\pi}{4}$ radians. The values printed into the serial monitor were read at each point, providing concrete values of p_{rad} and p_{sensor} . These values were put into the system of equations, and solving for this provided values for m and b :

$$m = -0.000359453251, b = 1.75789534 \quad (2)$$

3 Question D

The term 'stiction' describes the concept of static friction that exists in motors before they start moving. This creates a range of voltage values that, when sent to the motor, will not move the motor at all. This impacts the accuracy of the mechanism, especially when considering the fine movements required to balance a ball on a beam. This must be counteracted by creating a 'deadzone' at a certain range; for example, if the deadzone range is determined to be -0.1 to 0.1 volts, any voltage input within this range would be immediately rounded to either 0.1 or -0.1, depending on the direction.

The values of voltage that determine the deadzone range must be experimentally determined. Additionally, it is important to determine them in both directions, as the stiction in motors is not often symmetrical. In order to complete this task, a function was derived that, given a direction, methodically applies an increasing voltage to the motor, and only stops the loop once the potentiometer detects the motor start to change position.

The values for stiction have been determined for both directions, differentiated by whether the voltage values were increasing or decreasing:

$$v_{increasing} = 0.120V, v_{decreasing} = 0.101V \quad (3)$$

4 Question E

The objective of this question is to determine the equation which will convert the desired angular position into the necessary voltage that is to be sent to the motor. Said equation for a standard DC electric motor is as follows, with the full derivation in Appendix A:

$$\frac{\Theta(s)}{V(s)} = \frac{K_1}{s(\tau s + 1)} \quad (4)$$

While this is the regular transfer function for all standard DC motors, the constants K_1 and τ represent values that are different to each motor, including the resistance and inductance of internal components. In order to determine these values, the motor must be experimentally characterized.

At this point in the project, the system exists in an open-loop state, as can be seen in Figure 1, where the plant equation is the transfer function 4 and the controller is a simple proportional value. This system does not include a feedback loop, and therefore is not stable enough to be characterized accurately. By providing unity feedback (via the potentiometer, to feed back the current angular position of the motor) the system becomes closed-loop, as can be seen in Figure 2. and the system can be characterized.

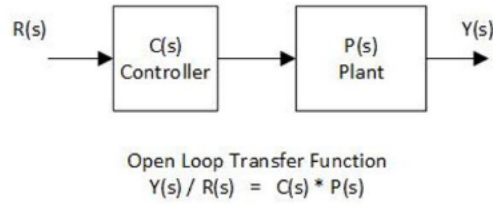


Figure 1: Open Loop System Block Diagram

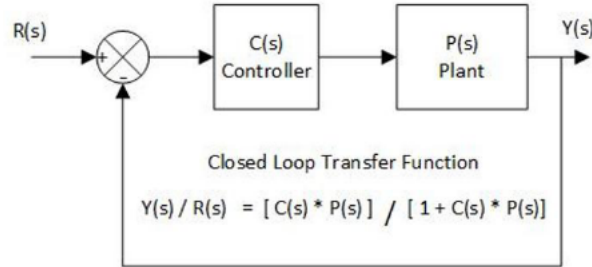


Figure 2: Closed Loop System Block Diagram

Due to the unity feedback, the transfer function changes to the following equation:

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{\frac{K_p K_1}{s(\tau s + 1)}}{1 + \frac{K_p K_1}{s(\tau s + 1)}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5)$$

The two new unknown values in the above equations, ω_n and ζ , can be determined through the following equations, which are based on the motor response when following a target theta value:

$$\zeta = \frac{-\ln(\frac{\%OS}{100})}{\sqrt{\pi^2 + (\ln(\frac{\%OS}{100}))^2}} \quad (6)$$

where $\%OS$ represents overshoot percentage - the percentage that the motor response overshoots the target signal;

$$\omega_n = \frac{\pi}{T_p \sqrt{1 - \zeta^2}} \quad (7)$$

where T_p represents the peak time - the time the motor response takes to reach its peak;

$$T_s = \frac{4}{\zeta \omega_n} \quad (8)$$

where T_s represents settling time - the time the motor response takes to reach 2% of the target value.

In order to determine the above values experimentally, a step response was created, and then input into the closed loop system as the target theta value. The error would then be calculated based on the current measured theta value and would be fed to the controller. Multiple values for the proportional coefficient of the controller were used, all of which provided a sufficiently large overshoot to measure; this was done to ensure experimental accuracy. The first value found to display a sufficiently large overshoot was $K_{p1} = -26$, and the next two values increased in steps of two, for $K_{p2} = -28$ and $K_{p3} = -30$

While a larger proportional coefficient value would have resulted in a larger overshoot, which would have been easier to study and use in the characterization process of the motor, the maximum value that should have been used in this case is $K_p = -30$. The lab manual has set two values that must be followed: the maximum error for this characterization exercise, which is $0.2rad$, and the maximum allowed voltage to be applied to the motor, which is $6V$. Since the voltage applied to the motor is determined by the controller, which multiplies the maximum error by the proportional coefficient, the highest proportional coefficient that would not exceed the maximum voltage is $K_p = -30$. Exceeding this limit would lead to saturation of the motor voltage in the code, which would result in an inaccurate characterization of the motor.

The values of the step response were sampled at a rate of 2000 times per second, or 2 kHz. This value was experimentally determined to be the most ideal for the given situation. The clock frequency was analyzed and was determined to be 48 MHz; given this value, as well as a rough estimate for how much space the step response sampling code would take up (roughly 4800 instructions), the ideal sampling rate was determined to be 10kHz. The interrupt service routine (ISR), which runs the sampling code, was set to run at this frequency. While this initially worked, noise from the ADC caused many outliers when measuring the motor step response. For this reason, the ADC was to be sampled five times and averaged for each value. This required the frequency to be divided by five (due to the increased rate of ADC sampling), resulting in a final sampling rate of 2 kHz.

The graphs for the step response of the motor can be seen below, in figures 3, 4, and 5.

From these graphs, the overshoot percentage and peak time values were determined to be as followed:

$$\%OS_{k=-25} = 12.415\%, T_{pk=-25} = 0.083069s \quad (9)$$

$$\%OS_{k=-28} = 15.47\%, T_{pk=-28} = 0.06758s \quad (10)$$

Motor Step Response ($k=-25$)

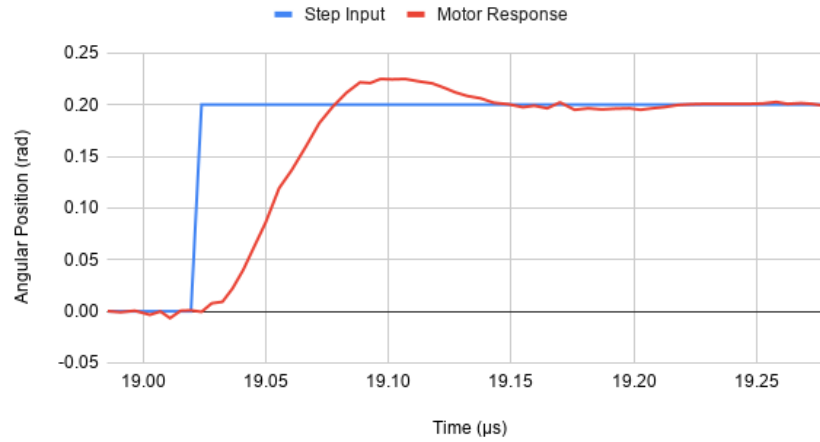


Figure 3: Motor response to step input when $K_p = -25$

Motor Step Response ($k=-28$)

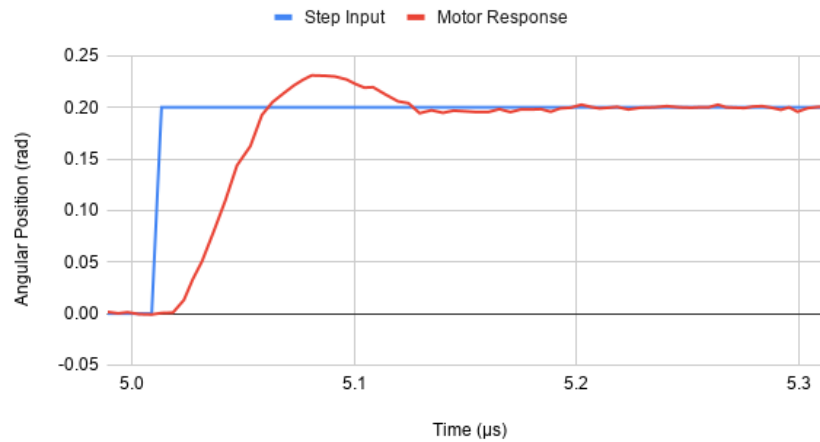


Figure 4: Motor response to step input when $K_p = -28$

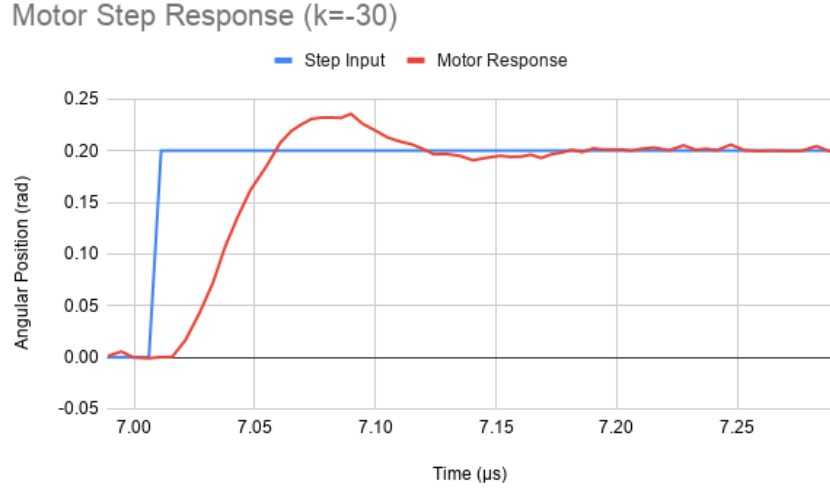


Figure 5: Motor response to step input when $K_p = -30$

$$\%OS_{k=-30} = 16.01\%, T_{p_{k=-30}} = 0.066472s \quad (11)$$

Using the formulas 6 and subsequently 7, the following values were able to be formulated:

$$\zeta_{k=-25} = 0.5532, \omega_{n_{k=-25}} = 45.3987 \text{ rad/s} \quad (12)$$

$$\zeta_{k=-28} = 0.5107, \omega_{n_{k=-28}} = 54.0710 \text{ rad/s} \quad (13)$$

$$\zeta_{k=-30} = 0.5037, \omega_{n_{k=-30}} = 54.7104 \text{ rad/s} \quad (14)$$

Substituting the above values into formula 5 gives the following K_1 and τ values for each K_p :

$$K_{1_{k=-25}} = -1.6413, \quad \tau_{k=-25} = 0.019909s \quad (15)$$

$$K_{1_{k=-28}} = -1.8906, \quad \tau_{k=-28} = 0.018107s \quad (16)$$

$$K_{1_{k=-30}} = -1.8103, \quad \tau_{k=-30} = 0.018144s \quad (17)$$

The values for K_1 and τ all fall within the ranges prescribed by the lab manual; with the motor constants properly defined, the motor has successfully been characterized.

5 Question F

Choosing $K_p = -28$ as an arbitrary selection, as well as its associated K_1 and τ values, the closed loop system used to characterize the motor was modelled in Simulink software. This was done to ensure that the characterization was done correctly, as accurate values for K_1 and τ would show a motor response which correctly follows $\theta_{desired}$.

A saturator function was also added to the modelled system in order to keep $\theta_{desired}$ within a given range, as in the future lab sessions, the beam will be attached to the motor, and moving

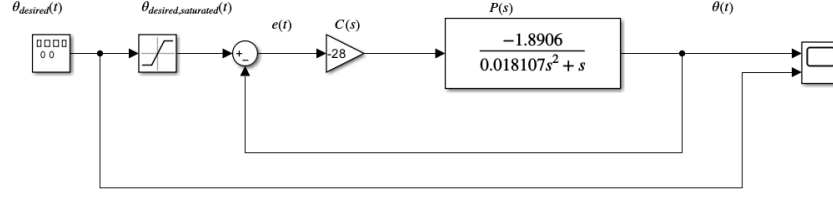


Figure 6: Closed-loop system modelled in Simulink software

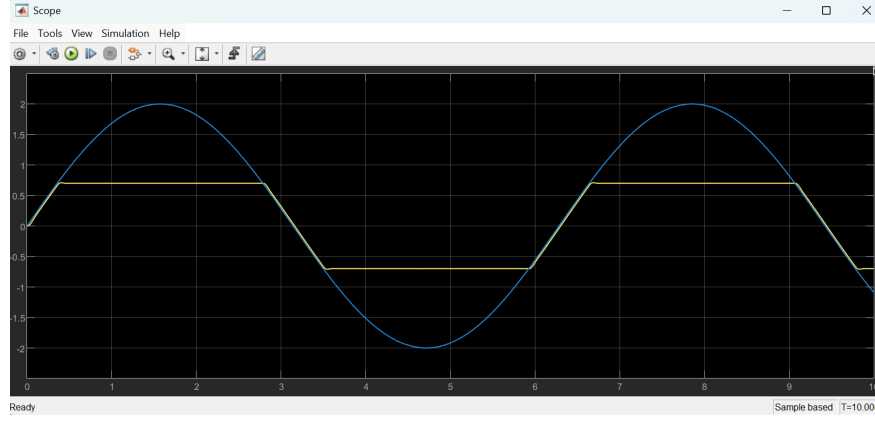


Figure 7: Closed-loop system model output; the blue line represents $\theta_{desired}$ and the yellow line represents θ

outside of the range may result in property damage. This saturator was also tested in the modelled system.

The code implementation for the target angular position saturator was quite straightforward. The saturation test results appear exactly as intended, with the angular position of the motor staying within 0.7 and -0.7, regardless of the value of the target theta value.

6 Appendices

6.1 Appendix A: Standard DC Electric Motor Transfer Function Derivation

Theta Saturation Test

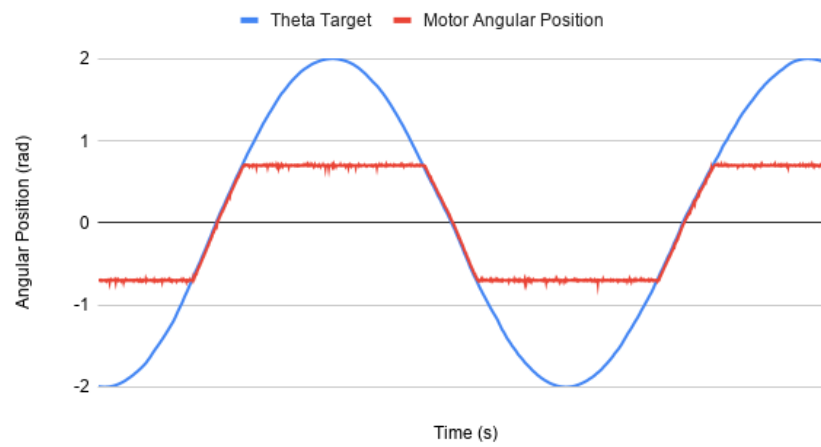


Figure 8: Theta Saturation Test

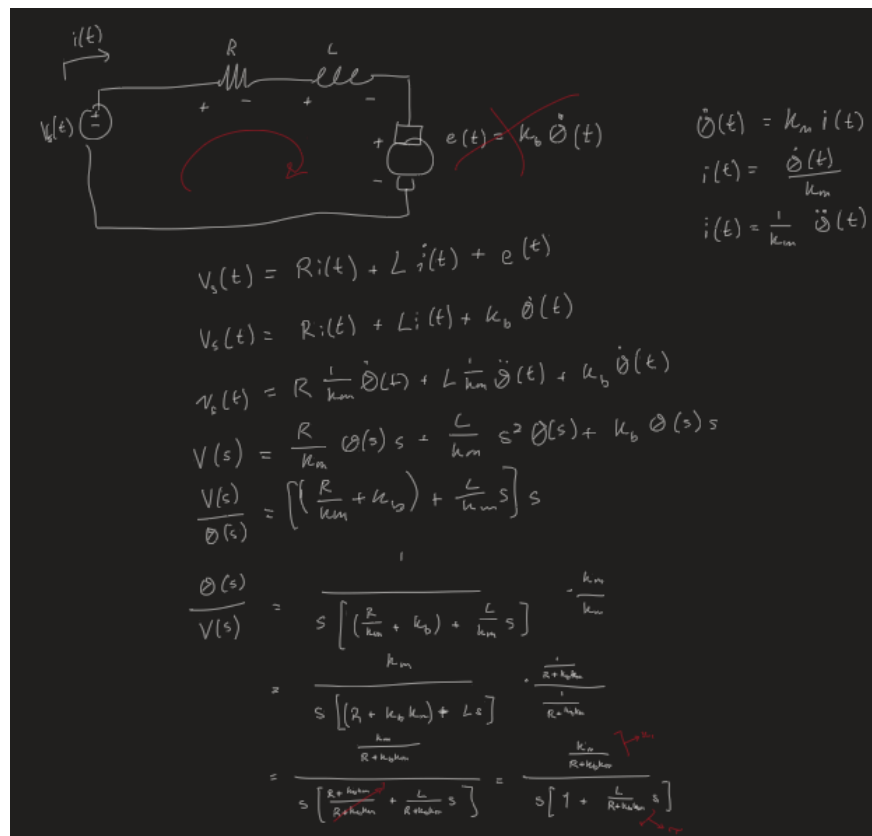


Figure 9: Standard DC Electric Motor Transfer Function Derivation